

Walter Wellness App

Jugal Patel (100744586)

Jerico Robles (1008776635)

Nadman Khan (100785940)

William Rory Chislom (100560820)

Ryan Fan (100876454)

Ontario Tech University

Human-Computer Interaction (CSCI 4620U)

Ali Neshati

Nov 6th, 2025

Table of Contents:

Abstract.....	2
Conceptual Design.....	4
Overview.....	4
Functions.....	4
Requirements.....	4
Non-essential features and presentation.....	5
Functionality.....	6
1. User Account Setup and Onboarding.....	6
2. Daily Check-In System.....	6
3. Mood and Habit Tracking.....	6
4. AI Generated Feedback and Insights.....	6
5. Progress Visualization Dashboard.....	7
6. Personalized Goal and Reminder System.....	7
7. Data Export and Professional Review.....	7
8. Privacy and Security Controls.....	7
9. Technical Performance and Accessibility.....	7
Table 1- Summary of all Key Functionalities.....	8
Scenarios.....	9
1. Story Scenario (based on p2b scenario 1):.....	9
2. Concrete Scenario (based on p2b scenario 2):.....	9
Interaction Design.....	11
Interaction Overview.....	11
Hierarchical Task Analysis.....	12
Interaction Flow Explanation:.....	13
1. Entry Point:.....	13
2. Primary Action:.....	13
3. AI Dialogue:.....	13
4. Progress Feedback:.....	13
5. Flow Continuity:.....	13
Design Consideration.....	13
Cognitive Load Reduction:.....	13
Emotional Tone:.....	13
Consistency and Predictability:.....	13
Accessibility:.....	13
Error Prevention:.....	14
Feedback Loop:.....	14
Interaction Model Summary.....	14
Design Sketches.....	15
Appendix.....	17

Abstract

This phase of the Walter Wellness project focuses on translating our earlier research findings into a concrete conceptual design, functional specification, and scenario based exploration of how users will interact with the system. Building on the insights gathered in Project Part 2 through questionnaires, semi-structured interviews, and task-based observations. The team refined a design direction centered on accessibility, trust, and supportive AI-driven wellness guidance. Participants from our earlier studies emphasized the need for simple onboarding, non judgmental emotional support, transparent data practices, and customizable engagement. These needs directly shaped the conceptual framework and proposed system behaviours outlined in this phase.

The design presented in Part 3a defines the core concepts of Walter Wellness as an AI powered mobile wellness companion that helps users track mood, sleep, habits, and daily reflections while receiving personalized feedback. We specify the key functionality required to support these goals, including adaptive mood-tracking interfaces, journaling prompts, flexible goal-setting, privacy controls, and integration with existing health APIs. Two detailed scenarios based on our persona Brandon illustrate how primary users with demanding schedules and limited access to healthcare may use WalterWellness to monitor their well being and develop sustainable habits.

Finally, this phase introduces an interaction focused task analysis demonstrating how users will complete essential actions within the proposed design. Together, these components form a research-informed foundation for the system's upcoming prototypes and interface sketches. This submission ensures that all design decisions remain grounded in user needs, stakeholder expectations, and the ethical considerations identified during earlier research stages.

Conceptual Design

Overview

The high-level concept for Walter Wellness is an AI-powered wellness app with strong personalization thanks to user-provided data and personal goals provided by the user. The main problem we are trying to solve (or mission statement) is having accessible, convenient healthcare support for people that need a low barrier to entry. Walter Wellness uses a companion metaphor to facilitate familiarity and help the user feel more at ease with a mascot, Walter, which is important for a wellness app. In order to ensure the needs of stakeholders are met and to help our solution for the problem of accessible healthcare, the plan is for the app to be free with easy setup and onramp.

Functions

The most important functions of the app are as follows: the ability to receive data tracked by the user's phone (partly integrated to their health app), a daily check-in system, receiving feedback from AI integration from data aggregated by health app integration and user-provided information, and notifications for long-term goals and health journalling. More secondary functions of the app include: a seamless onboarding and setup process, mood and habit tracking, progress visualization, data export, privacy/security controls, motivation and encouragement support, and sufficient technical performance and accessibility features.

The core of the app is its ability to receive user data from information that's already tracked as well as receive data entered by the user themselves, as it's what enables AI to create suggestions and assessments of the user's well-being. This allows the basic feedback loop to interact with the app, satisfying stakeholder/user needs for personalization and branches off further with additional functions. For the AI to perform its analysis it needs information over a long period of time which is supplemented by a check-in system to encourage engagement with the app, along with notifications based on an individual user's goals and needs. Mood and habit tracking, progress visualization, and data export functionality would let the user see information about their own health over a long period of time, which can be combined with more standard brick and mortar/in-person methods of healthcare to create plans that will work for the user. Privacy controls help the user feel more safe using the app, along with motivation and encouragement support helping the user feel comfortable and personalized. Seamless onboarding and setup, sufficient technical performance and accessibility features would minimize barriers to usage and offer important quality of life, which is important for user engagement.

Requirements

The most important stakeholder is the user, and we based much of our research on the feedback that would-be users from our target demographic (adults and young adults). From there, the most important requirements we gathered from that are easy account setup, personalization, monetarily free/accessible, long term goals and progress tracking, quality

notifications, user transparency, measurable engagement metrics, and emotional support through motivation/encouragement. All of these are covered by planned functions. Requirements relevant to other stakeholders, like healthcare providers and public health organizations, are covered through data export to aid support provided by healthcare providers, and privacy and safety options to assist in ethical data collection.

Additional technical requirements that assist in our solution include: secure data encryption and local storage options, integration with existing health APIs, lightweight architecture, automatic data backup, and scalable backend with AI module isolation. Secure data encryption and local storage options, along with AI module isolation are relevant to user transparency and security, which are important to making a user feel comfortable. Integration with existing health APIs is important for app functionality. Lightweight architecture and automatic data backup are important for quality purposes and ensuring good user engagement for long-term use.

Non-essential features and presentation

Non-essential features of the app include a clean UI/UX, Walter existing as a metaphor to provide comfort to the user, and Walter voicing over text with cartoon gibberish speak.

Functionality

To list the capabilities of the system the Walter Wellness that will provide an insufficient detail that will be clear to someone that is not on our design team is that the:

1. User Account Setup and Onboarding

This will help users to create a secure account that will have minimal input to reduce the friction. The onboarding process will introduce the app's privacy policies and the data handling procedures in plain language, which will enable full transparency and explicit consent. The user will then personalize the notification with their preferences, and then select a wellness feature that will focus on the area such as mood, sleep, or energy), and then complete their initial check-in that will calibrate the AI assistant. This feature will then ensure accessibility for the new users while establishing the trust from the outset.

2. Daily Check-In System

The daily check in will allow the users to log their current emotional state with sleep quality, and overall energy level that will use the intuitive slides, emojis, or short text entries. The AI assistant will then immediately provide a brief reflection or supportive comments, that will reinforce the daily consistency and emotional awareness. This feature will help the user to establish a routine of mindful self reflection without the cognitive overload that is associated with the detailed journaling.

3. Mood and Habit Tracking

Are one of the functions that are beyond the basic checkins, where the user will record their habits such as exercise, journaling, or caffeine consumption to track behaviour patterns. The app will then convert these data into a mood timeline that visualizes emotional and physical trends across days or weeks. The purpose of this feature is to identify the patterns that may influence stress levels or motivation, providing both users and healthcare professionals with measurable insights.

4. AI Generated Feedback and Insights

The integrated AI engine analyzes the logged data that provides contextual feedback such as "You've been calmer this week after more consistent sleep." These insights would emphasize the encouragement and the progress rather than clinical assessment. The AI also suggests small, achievable actions such as breathing exercises or brief reflection prompts, and personal goals that are tailored to the user's recent patterns. Over time, the model adapts to user input, refining the tone and relevance of recommendations.

5. Progress Visualization Dashboard

Logs all the data that are clustered in an interactive dashboard that uses clean colour gradients, trend lines, and summary cards. The dashboard would then highlight the emotional fluctuations, sleep duration, and activity frequency. Unlike other traditional health graphs, the interface would have calm and reduce the anxiety to make the self tracking more approachable. This visualization will not only keep users more informed but also will provide the motivation by clearly showing gradual improvement.

6. Personalized Goal and Reminder System

Users may set the short term goals like logging the mood daily for one week or Reduce caffeine intake. The reminder system will then customize and allow a flexible frequency with a message tone. The notifications are then phrased positively to “How are you feeling today?” and can be muted or scheduled during a preferred hours. This system will support users to autonomy while maintaining a gentle encouragement for sustained engagement.

7. Data Export and Professional Review

For secondary stakeholders such as the therapists or doctors, the users can export their wellness summaries in PDF or CSV formats. These reports will then include the user’s mood trends, consistency scores, and AI generated summaries that will capture the high level patterns without revealing sensitive raw data. This functionality bridges the gap between digital self care and professional health review, giving the healthcare providers a more concise and ethical insights into the user’s wellness.

8. Privacy and Security Controls

The privacy dashboard will allow the users to view, manage, and delete the stored data at any time. The sensitive information is encrypted locally before the transmission, and all AI analytics can operate on anonymized datasets. Users can opt out of sharing data with third parties or healthcare providers. This will transparent the structure to ensure the compliance with ethical standards and build the trust between the user and the system.

9. Technical Performance and Accessibility

The app’s architecture is designed for the speed and reliability across the devices. It will integrate a lightweight AI model that is capable of running it locally or through secure cloud endpoints. The Accessibility features will include a high contrast theme, large font mode, and voice interaction for visually impaired users. These features will ensure inclusive usability for diverse audiences.

Table 1- Summary of all Key Functionalities

Functionality	Purpose	Stakeholder Benefits
Account setup & onboarding	Simple sign up and ensures data consent	All stakeholders
Daily Check In	Enables a quick emotional and physical tracking	Primary users
Mood and Habit Tracking	Visualize long term trends	Users, healthcare providers
AI Feedback and Insights	Provides motivational and adaptive responses	Users, developers
Progress Dashboard	Displays emotional and behavioral patterns	users
Goal and Reminder System	Encourages healthy routines	Users
Data Export	Share summarized reports	Healthcare providers
Privacy Dashboard	Control personal data usage	Users, ethics teams
Accessibility and Performance	Ensure smooth and inclusive user	All stakeholders

Therefore, the Walter Wellness App will integrate these functionalities to create a seamless user experience that is supportive, ethical, and adaptive. Each feature not only enhances individual wellness engagement but also strengthens trust through transparency and inclusive design. The system's functional scope that ensures that all stakeholder needs will be emotional, ethical, technical, and professional. Which addresses and forms the foundation for the interaction and the interface designs that follow.

Scenarios

1. Story Scenario (based on p2b scenario 1):

Monday morning Brandon gets up early as usual to go to his shift at McDonalds. He is anxious about the two projects that need to be completed by tonight and midterms on Wednesday and Friday.

On the bus to work, he is overwhelmed just thinking about the upcoming work that needs to be done. Then he remembered he downloaded the *Walter Wellness App*. He opens it and is greeted by Walter and prompts him: "Good morning, how are you feeling today?" Brandon enters "stressed" and types a short answer "worried about the projects and midterms". Based on the reply the AI quickly noticed his stress and provided a supportive response "You've got this! Remember to get breaks between your work". Then it suggests a breathing exercise to reduce his restlessness.

During the shift breaks, he got notification from the app: "Based on your schedule, after the shift you have a 45 minute block of time before your class starts. Would you like to set a goal to work on your project?" Brandon selects confirm and the app automatically sets the goal and reminder.

After the morning shift and the class, Brandon got a reminder from the app, first the app complimented his work throughout the day and fed him a Progress Visualization Dashboard to show he completed an assignment. It also shows the mood improvement from stressed to normal. The calm interface of the dashboard makes Brandon feel relieved and comfortable after a stressful day.

2. Concrete Scenario (based on p2b scenario 2):

Goal: Brandon wants to improve sleep and reduce the reliance on caffeine. And maintain energy throughout the week.

Context: Recently Brandon is not feeling so well and always has a cup of coffee in his hand. He realizes he has trouble sleeping and has a habit of using caffeine to help him go through the day.

Narrative: Brandon decides to use the app to rebuild his mood, sleeping cycle, and less reliance on caffeine.

- First he opens the app and quickly finishes the daily check in and then he goes straight to the mood and habit feature.

- The app pops up the new sleeping analysis table throughout the week and shows he stays up late and the sleep quality is low.
- Then the app prompts a long term goal suggestion based on the bad habit and sleep.
- Brandon accept the goal and the app adds a new tracker for the coffee intake and set a reasonable sleep time for his busy schedule
- For the next few weeks, once Brandon has a coffee he will tap on to the app and by the end of the day the app calculates how much caffeine he consumed and give support and methods to reduce the needs of coffee.
- Every morning when he wakes up, he will see a sleep result for how well he slept last night. And give suggestions to how to improve it.
- After a month Brandon got a notice from the app that congrats him and showing the improvement from last month and motivate him to keep up on drinking zero caffeine and keep on to the good sleep throughout the week

Key design feature:

- Active/passive tracking feature for calculating sleep and caffeine consumed.
- Goal setting
- AI analysis and suggestions: Based on user data to give the best advice
- Feedback for long term goals and tasks

Interaction Design

The interaction design of the Walter Wellness App will illustrate how the users will perform the essential tasks through an intuitive, step by step sequence of the interactions. It will focus on how people will accomplish their wellness related goals using the proposed system and how those tasks are simplified through a supportive AI responses. Unlike the task analysis in Phase 2b, which described the current process of tracking the wellness manually or through existing apps, this section will demonstrate how the same task can be accomplished using the Walter Wellness.

The key interaction that was chosen for analysis was the Daily Mood Logging and Reflection sequence where one of the app's most critical task was to capture how the users can engage with the system, how the AI can give feedback that can provide and interact how the flow will naturally go from one step to another.

Interaction Overview

The daily mood logging will process the beginning of when the user will open the Walter Wellness App. The system will greet the user with a calming interface that will immediately set a positive emotional tone. From there, the user will perform a quick self check in by selecting a mood or writing a short note about their current state. The AI will respond in real time with supportive comments and record the data for visualization in the dashboard. The overall interaction flow is designed to feel the conversational and non clinical way to emphasize the small effort of immediate feedback, and emotional safety that encourages a consistent, long term engagement.

Hierarchical Task Analysis

The goal is to complete a Daily Mood Log and Receive a Personalized AI Feedback

Task Step	User Action	System Response	Purpose or Design Rationale
1. Launch the App	Taps the Walter Wellness Icon	Loads the home screen with a gentle animation and greeting by saying Hi <user>, how are you feeling today?	Creates a positive emotional context and lowers stress
2. Begin Daily Check In	Taps the Start Check In button	Displays the mood scale with the emoji slider or numeric scale	Simplifies the process to reduce the cognitive load
3. Select Mood or Energy Level	Choose from calm to stressed scale, that optionally adds the short text note	Stores the selection and confirms the input visually	Makes the input quick and visually satisfying
4. AI Reflection and Response	Wait for system to process entry	Generates a contextual motivational message	Provides instant feedback and emotional reinforcement
5. Optional Add on Details	Tap the "Add Activity" or "Sleep Quality"	Expands the secondary input cards	Allows the flexibility without forcing a long input.
6. Save Entry	Press "Save"	Confirms a success with message ("Entry saved! See your trend on the dashboard")	Reinforces task completion and success.
7. Review Progress	Optionally tap "View Dashboard"	Displays graphs of mood trends, sleep, and reflections	Encourages continued use through visual improvements

Interaction Flow Explanation:

1. Entry Point:

The user's interaction will begin from the home screen and greet the user by their name and interact in a calm tone to immediately establish a personalized and emotional connection.

2. Primary Action:

The main interaction involves selecting a mood. By using a colour coded scale with the emojis, the user can express their emotional state quickly without typing.

3. AI Dialogue:

After the input submission, the AI component will generate short, empathetic reflections. This will create a sense of conversation rather than data entry.

4. Progress Feedback:

Each interaction will lead to a visual confirmation with either a Success message or a small animation for reinforcing progress.

5. Flow Continuity:

The design will avoid the transitions with every interaction that will end with a clear next step such as (Save, View Dashboard, or Set Goal), by maintaining the engagement and reducing the user drop off.

Design Consideration

Cognitive Load Reduction:

- Each screen will focus on one clear task to prevent overwhelmed users who may be stressed or fatigued

Emotional Tone:

- The language used by the AI will be supportive, motivational, and free from judgment.

Consistency and Predictability:

- Icons, colours, and animations will follow a calming aesthetic to ensure a consistent visual communication.

Accessibility:

- The interaction can be completed entirely through touch, with optional voice input for accessibility.

Error Prevention:

- The system will include confirmation prompts and undo options to prevent accidental submissions.

Feedback Loop:

- Every user action will produce an immediate, visible response to confirm that the system has received their input.

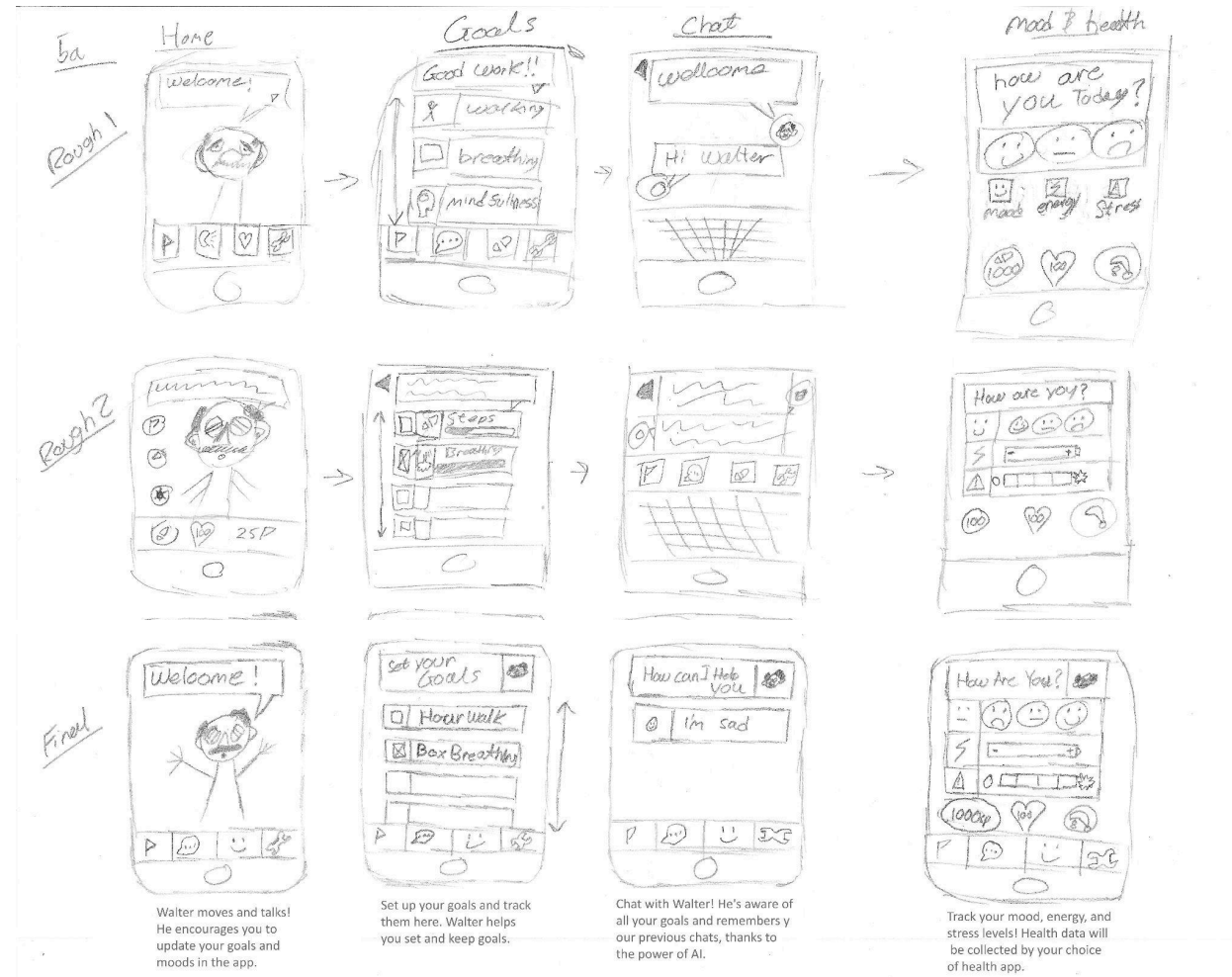
Interaction Model Summary

Interaction Aspect	Design Focus	Intended Effect
Input	Simplified one tap selection	Encourages daily use
Feedback	Immediate AI reflection	Builds emotional connection
Control	User can skip or customize options	Enhances comfort and autonomy
Context	Consistent tone and colour theme	Creates a safe, calming environment
Adaptability	Learns from past entries	Increases the personalization and trust

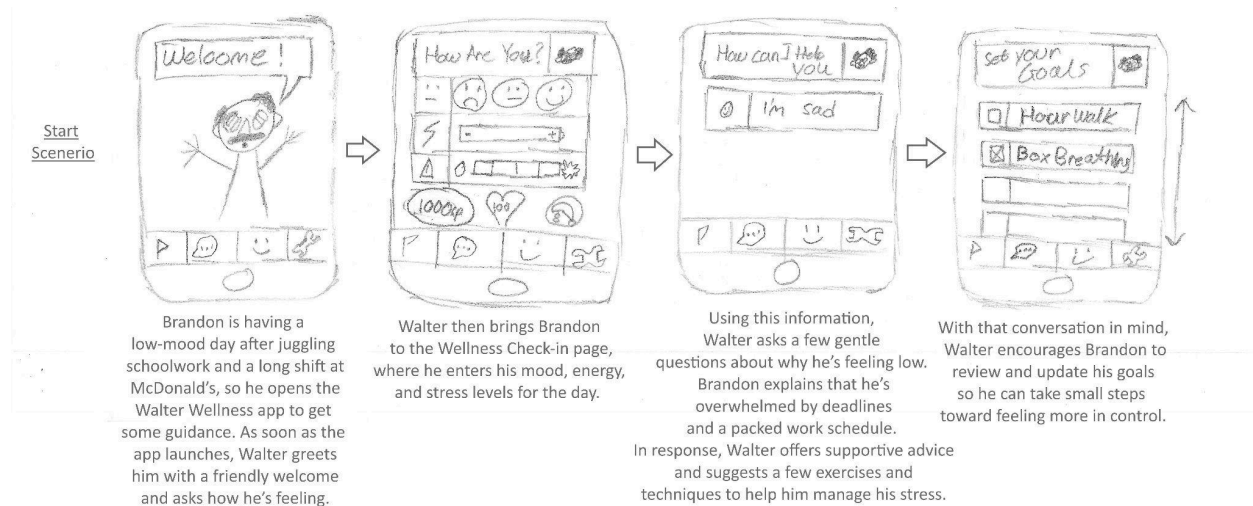
This interaction sequence will demonstrate how the users can effortlessly integrate Walter Wellness into their daily routine. The hierarchical task analysis will show a logical, empathetic flow that supports the wellness reflection without requiring significant time or effort. Each design choice from the input simplicity to AI feedback will enforce the app’s core values of accessibility, emotional support, and user trust. By aligning technology with human centered interaction patterns, Walter Wellness will transform the stress management from a passive tracking experience into an active, motivating dialogue that fosters a long term engagement.

Design Sketches

Interface Design



Interaction Design



Appendix

Jugal Patel:

- Carried out each phases of the project and organized the word doc
- Kept everyone on track for submission
- Used discord to communicate with group members
- Group leader for the final project
- Ensuring everyone gets their work done
- Submitted most of the phases by the deadline.

William Rory Chislom:

- Set up the initial doc for part 3a
- Led meetings
- Assigned work for part 3a
- Did the design sketches
- Proofread other group members' work

Jerico Robles:

- Made first draft of the project idea with Nadman
- Discussed project plan within and after class in meetings
- Covered assigned roles for Nadman when he couldn't get his tasks done
- Contributed Jugal and Rory with getting tasks done on time before the deadline
- Proofread other group member's work
- Did the conceptual design part of this document
- Submitted phase 2b by the deadline

Nadman Khan:

- Contributed by helping Jugal set up the Google Doc and organizing it
- Assisted with meeting the project requirements
- Gave ideas to Jugal for most of the phases
- Did the abstract for most of the phases.
- Assisted Rory with Group Meetings through discord

Ryan Fann:

- Participated in group meetings
- Worked on the persona and scenarios for part 2b
- Worked on the scenarios on storyboarding for part 3a
- Proofread other group member's work